

Experiment:37

Construct a C program to simulate the First Come First Served disk scheduling algorithm.

Code:-

```
#include <stdio.h>

#include <stdlib.h>

int main()
{
    int n, head, seek_time = 0;

    printf("Enter the number of disk requests: ");

    scanf("%d", &n);

    int request_queue[n];

    printf("Enter the disk request queue:\n")

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &request_queue[i]);
    }

    printf("Enter the initial position of the disk head: ");

    scanf("%d", &head);

    printf("\nFCFS Disk Scheduling:\n");

    printf("Head Movement Sequence: %d", head);

    for (int i = 0; i < n; i++)
    {
        seek_time += abs(head - request_queue[i]);

        head = request_queue[i];

        printf(" -> %d", head);
    }
}
```

```
printf("\nTotal Seek Time: %d\n", seek_time);  
printf("Average Seek Time: %.2f\n",  
      (float)seek_time / n);  
return 0;  
}
```

Output:

```
Enter the number of disk requests: 6  
Enter the disk request queue:  
12 24 48 816 6 1632  
Enter the initial position of the disk head: 6  
  
FCFS Disk Scheduling:  
Head Movement Sequence: 6 -> 12 -> 24 -> 48 -> 816 -> 6 -> 1632  
Total Seek Time: 3246  
Average Seek Time: 541.00  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```